Day 8 Data Report

Multi-sensor self-monitoring, 15 October 2026,
with comparison to days 1–7

Participant: Hemel

Analysis generated with `pipeline/daily_analysis.py`,
`compare_days.py`, `relationships.py` and
`walk_hydration.py`

1 Summary

Day 8 was a follow-up day with one purpose: to test again whether drinking water before a walk lowers heart rate while walking. The participant drank 500 mL shortly before a 13-minute morning walk (11:30–11:43) recorded with both insoles, the environment node and GPS; the wrist band recorded the whole day until 15:52. Table 1 lists the headline metrics. The main findings are:

- **Water before walking: not supported.** Despite the 500 mL, walking heart rate rose from about 86 bpm at rest to a median of 123 bpm (mean 120 bpm excluding suspected cadence lock), the *largest* rise above rest (+37 bpm) of all eleven walks of the study. With day 8 added, the water drunk in the 2 hours before a walk is unrelated to walking heart rate ($\rho = 0.20$, 95 % CI −0.56 to 0.79) and to its rise above rest ($\rho = -0.31$, CI −0.86 to 0.57; after allowing for time of day +0.04). The low heart rate on the day-7 evening walk was not reproduced; the earlier pattern is explained by the time of day (Section 6).
- **Gait** was fast and symmetric: cadence 108 steps/min (the highest of the study), stride 1.11 s, speed 1.25 m/s and stride length 1.37 m; stance 61.8 vs 62.6 % of the stride (left vs right), push-off and step timing symmetric (572 paired strides). All six pressure sensors were healthy: the right heel sensor that failed on day 7 worked normally again.
- **Sleep** was split into two parts, 02:08–05:28 and 07:45–10:53, with 2.3 h awake in between: 6.3 h asleep in total.
- **Heart rate and step rate** remained positively related on all eight days (pooled $r = 0.30$; sign test $p = 0.008$).

Table 1: Headline metrics of day 8 (15 October 2026).

Domain	Metric	Value	Unit
Sleep	Time asleep	6.3	h
	Sleep efficiency	98	%
	Light / deep / REM	50 / 27 / 23	%
	Mean HR asleep	64	bpm
	Mean SpO_2 asleep	96.9	%
Activity	Steps (band / phone)	2,347 / –	
	Mean awake HR	86	bpm
	Minutes with HR $>$ 120 bpm	10	min
	HR–steps Spearman ρ	0.34	
	HR lag behind steps	7	min
Gait (walk)	Cadence	109	steps/min
	Stride time / CV	1.10 / 7.9	s / %
	Stance, IMU (R / L)	62.6 / 61.9	%
	Swing, IMU (R / L)	0.416 / 0.425	s
	Stance, pressure (R)	–	%
	Push-off timing (R / L)	65.4 / 66.2	%
	Distance / speed	741 / 1.25	m / m/s
	Stride length	1.37	m
	Insole − band bias	-2.3	steps/min
Posture	Monitored time	–	min
	Upright / slouched	–	%
	Median time between changes	–	min
Environment	Indoor CO_2 (median)	–	ppm
	Indoor / outdoor $PM_{2.5}$	– / 59	$\mu g/m^3$
	Indoor temp. / RH	–	°C / %
	Indoor / outdoor sound	– / 70	dBA
	Bedroom CO_2 / $PM_{2.5}$	–	ppm / $\mu g/m^3$
	Bedroom temp. / sound	–	°C / dBA
Left/right	Left lag behind right	50	% of cycle
	Foot-tilt range L − R	-1.4	°
	Stance / swing SI (IMU)	-0.3 / +2.1	%
	Stance % (L / R, IMU)	61.8 / 62.6	%
Hydration	Total intake (drinks)	–	mL
	Longest gap between drinks	–	h

2 Data collected

Table 2: Recording on day 8 (local time, UTC+6).

Key	Session	Time	Sensors	Notes
W	Morning walk	11:30–11:43	Insoles, environment, GPS	500 mL drunk shortly before
–	Wrist band	00:00–15:52	Heart rate, steps, sleep, SpO_2	Export ends at 15:52

The upload contained the walk's files without a session folder; the session times were taken from its event log. The exact time of the drink was not logged; it is entered as 11:20, and any time within the 2 hours before the walk gives the same analysis. No other drinks were reported, so no daily fluid total is given. The band's export ends at 15:52, so the day's step count (2,347) covers only part of the day. Temperatures include the −5 °C correction; the right first-metatarsal sensor is still the replacement type, so stance and swing come from IMU gait events.

3 The morning walk

The participant walked 741 m in 11 minutes at a steady 1.25 m/s (Fig. 1). The right insole detected 625 and the left 618 strides: median stride time 1.11 s, cadence 108 steps/min and stride length 1.37 m, the fastest walk of the study, with low variability (stride-time CV 7.9 %; Figs. 3 and 4). The outdoor air had $PM_{2.5}$ of 59 $\mu g/m^3$, 31 °C and 70 dBA (Fig. 5).

Heart rate during the walk (Fig. 2). While sitting before the walk heart rate was 61–75 bpm (10:30–11:08); it rose to 83–100 bpm after the participant got up at about 11:10, and during the walk it climbed steadily from 103 to 127 bpm. Three walking minutes (11:33–11:35) were flagged as possible cadence lock (heart rate within 6 bpm of the step rate); excluding them the walking mean was 120 bpm, in the middle of the participant's range on other walks (111–124 bpm). After the walk heart rate stayed at 105–127 bpm for about 25 minutes.

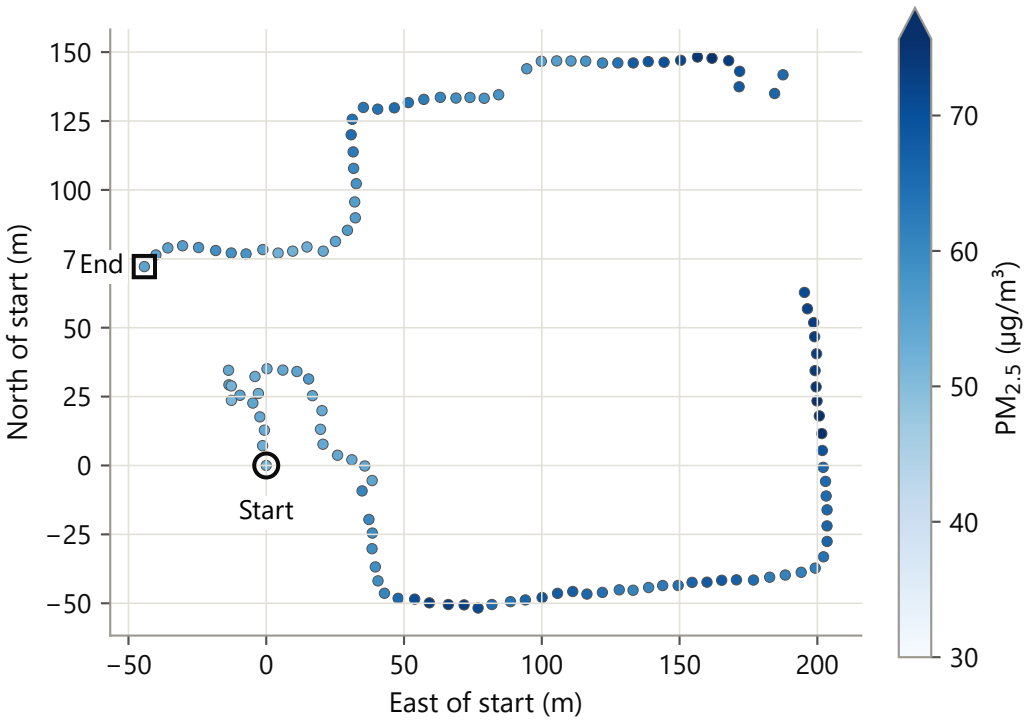


Figure 1: GPS track of session W (fixes with accuracy $\leq$20 m; logging began at 11:31): 741 m in 11 min at a median speed of 1.25 m/s, in metres from the start point, coloured by the $PM_{2.5}$ concentration measured by the body-worn environment node (colour scale capped at the 98th percentile, 76 μg/m^3; maximum 78).

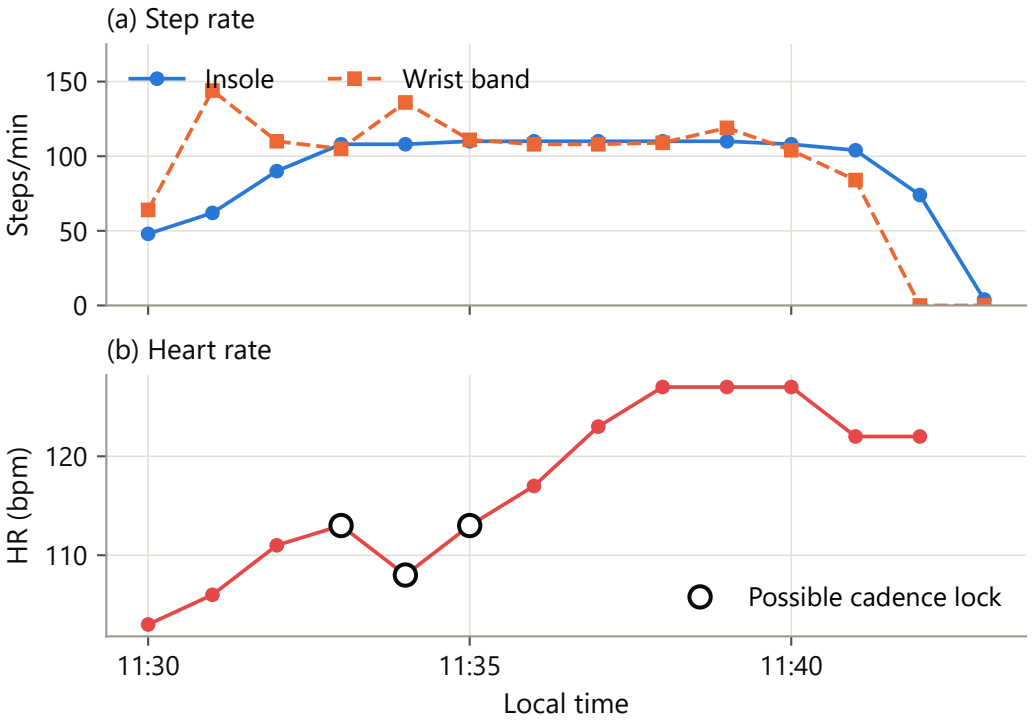


Figure 2: Session W (morning walk after drinking 500 ml) minute by minute: (a) step rate from the insoles (2 × the strides of whichever insole detected more in that minute) and the wrist band; (b) wrist-band heart rate (mean 117 bpm). Open circles: 3 minutes in which the band's heart rate was within 6 bpm of the step rate for at least two consecutive minutes, a sign that the optical sensor may have locked onto the walking rhythm instead of the pulse.

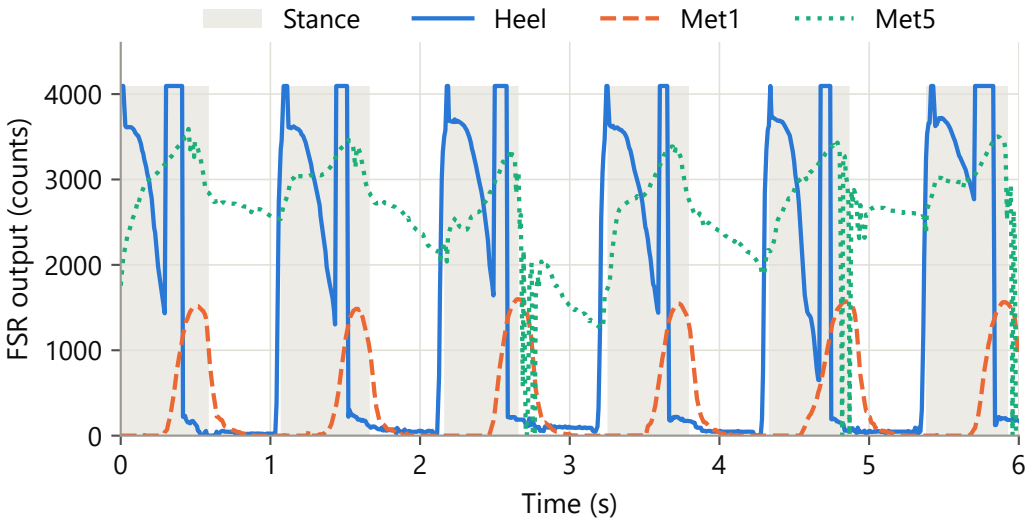


Figure 3: Right-insole plantar pressure during 6 s of steady walking (session W, morning walk after drinking 500 ml). Lines show the heel, first-metatarsal (Met1) and fifth-metatarsal (Met5) force-sensitive resistors (12-bit ADC counts; higher means more load). Shaded intervals are stance phases detected by hysteresis thresholding of the summed, 6 Hz low-pass filtered load.

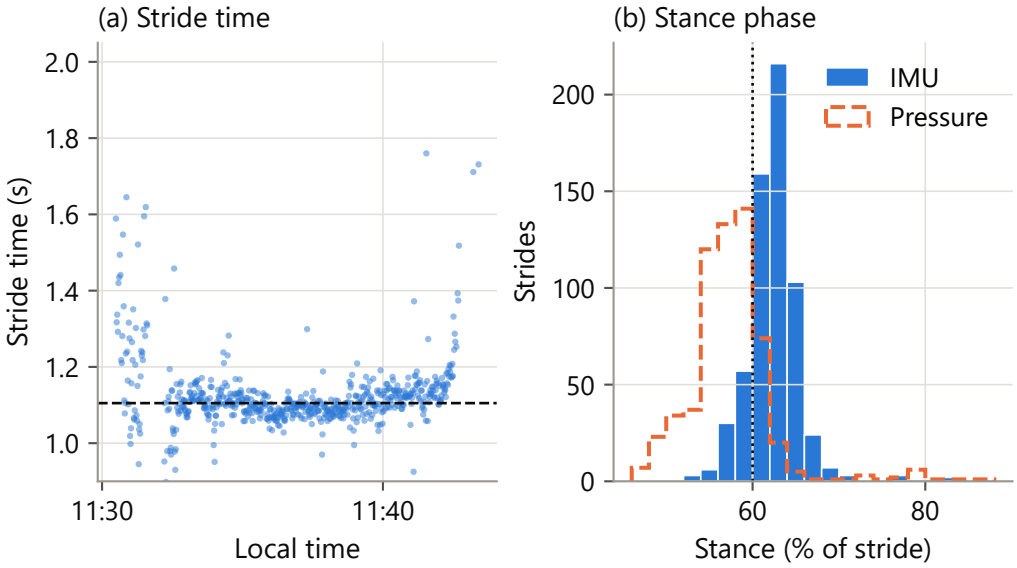


Figure 4: Right-foot stride parameters, session W (625 strides). (a) Stride time of every detected stride; dashed line = median (1.11 s, cadence 109 steps/min). (b) Stance phase as a percentage of stride time from IMU gait events (bars; heel-strike impact to IMU toe-off; median of steady-walking strides 62.6 %) and from the pressure sensors (dashed outline; 57.5 %) – not valid on this day because the first-metatarsal sensor had been replaced; dotted line = typical healthy value of about 60 %.

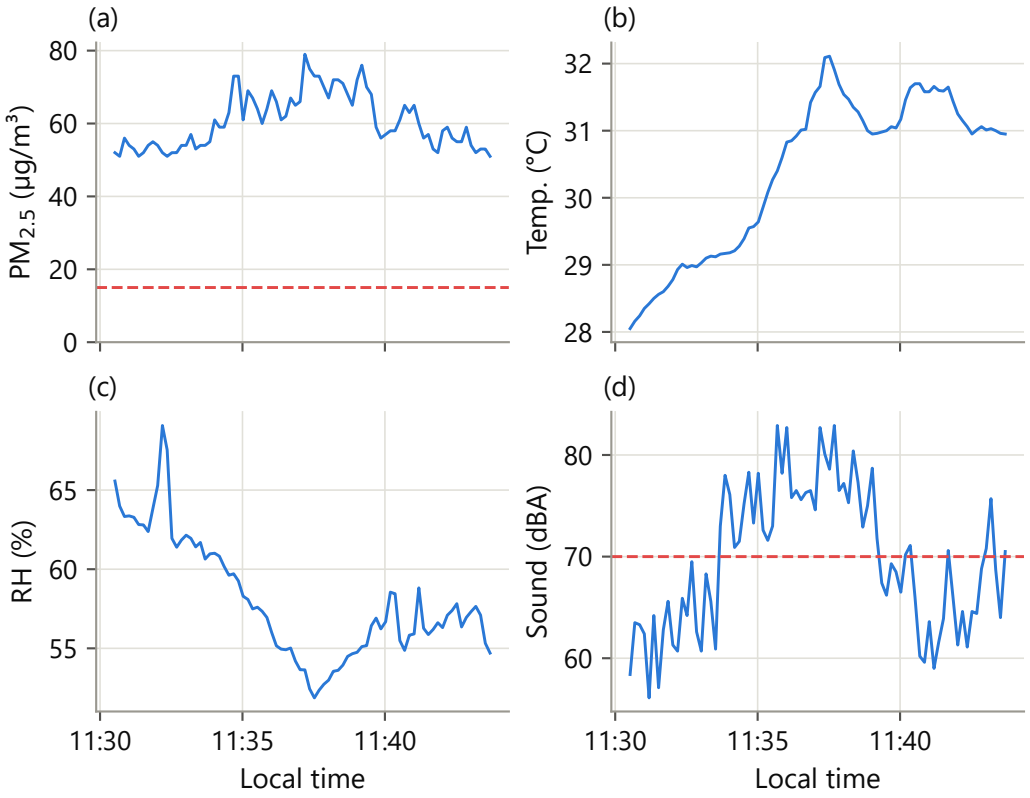


Figure 5: Environment node during session W (morning walk after drinking 500 ml, 10 s averages): (a) $PM_{2.5}$, median 59 $\mu g/m^3$, peak 79 (dashed: WHO 24-h guideline); (b) temperature, median 31.0 °C; (c) relative humidity, median 57 %; (d) sound level, median 70 dBA (dashed: 70 dBA). CO_2 is omitted because readings fell below the physically plausible outdoor level of about 420 ppm (minimum 128 ppm). Temperature includes a -5 °C correction for sensor self-heating.

4 Left versus right foot

Both insoles streamed at 112–114 samples/s and all six pressure sensors were healthy. 572 strides were paired (Fig. 7): stride time +0.1 %, stance time −0.3 % (0.687 vs 0.689 s), swing time +2.1 %, stance 61.8 vs 62.6 % of the stride, push-off timing +1.2 % and foot-tilt range −2.4 %, all within the ±5 % band; step timing from heel-strike impacts was 49.4 % of the cycle (Fig. 8). With the right heel sensor working again, the foot-tilt range was again slightly smaller on the left, as on days 1, 3, 5 and 6.

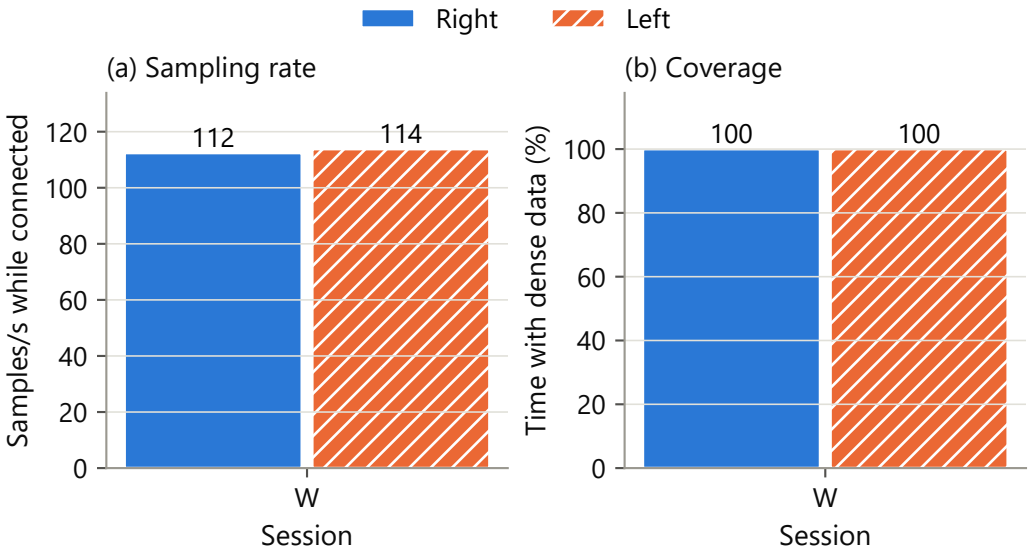


Figure 6: Data delivered by the right and left insoles in each session with both feet: (a) average sampling rate while the node was connected (nominal about 112 Hz) and (b) share of the recording covered by densely sampled data (consecutive samples ≤20 ms apart). The left insole's low coverage limits the left/right comparison to cycle-averaged timing and shape.

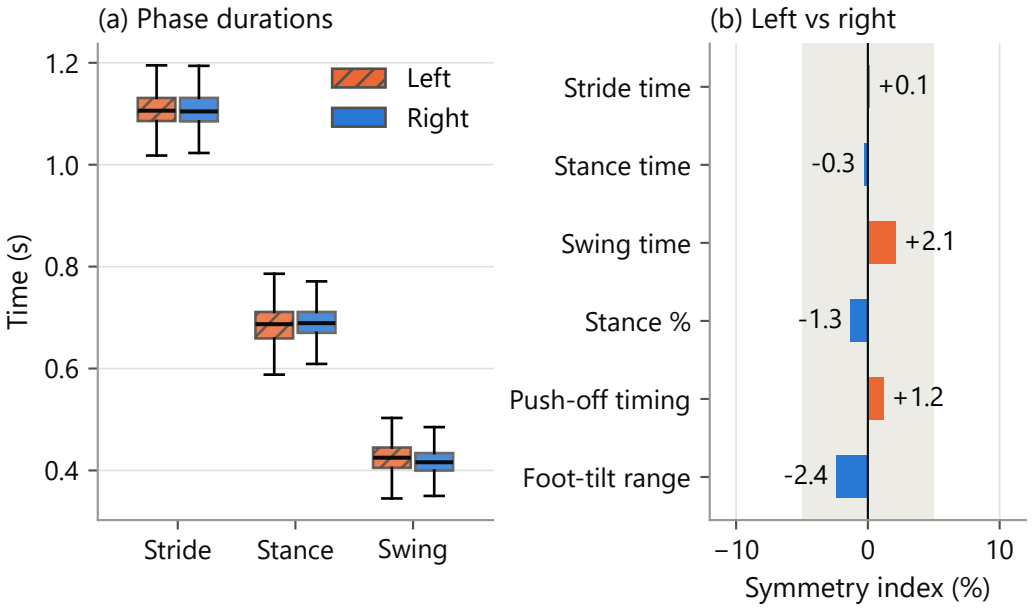


Figure 7: Stride-level comparison of the left and right foot during session W (572 paired strides: each right stride with the left stride that begins inside it; strides deviating more than 15 % from the local median were excluded as detection errors). (a) Stride, stance and swing durations per foot. (b) Symmetry index SI $= 100\,(L - R)/\overline{LR}$ of the medians (positive: larger on the left); the grey band marks ±5 %. Stance and swing are bounded by IMU gait events (heel-strike impact to IMU toe-off), defined identically on both feet and independent of the pressure sensors. Medians, left vs right: stride 1.106 vs 1.105 s, stance 0.687 vs 0.689 s (61.8 vs 62.6 % of stride), swing 0.425 vs 0.416 s, foot-tilt range 55° vs 56°. Pressure-based stance for comparison: 60.6 vs 57.4 % (right not valid: first-metatarsal sensor replaced by a different type).

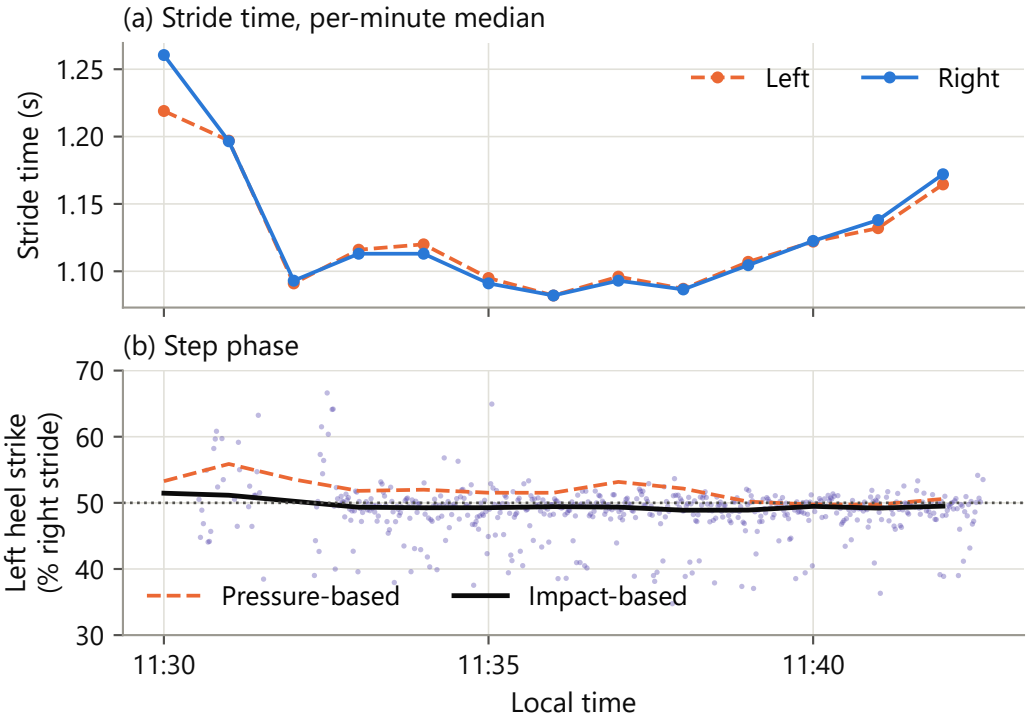


Figure 8: Left and right foot over session W. (a) Per-minute median stride time of each foot; the curves coincide, confirming that the two independently timed insoles agree (paired-stride correlation $r = 0.80$). (b) Timing of each left heel strike within the concurrent right stride with heel strikes taken from the accelerometer impact (dots and solid line: per-minute median; overall 49 %) or from the pressure sensors (dashed; overall 52 %); 50 % is symmetric. The pressure-based estimate drifts because the delay between pressure-detected heel strike and impact changed during the session on the left insole (28, 27, 28 ms in successive 5-min blocks; right: 0, 0, 18 ms). Clock drift is corrected; a constant radio-latency difference of a few tens of milliseconds between the nodes cannot be excluded.

5 Sleep and the rest of the day

The band recorded two sleep periods (Fig. 9): 02:08–05:28 and 07:45–10:53, with 2.3 h awake in between, 6.3 h asleep in total (27 % deep, 23 % REM). Mean heart rate asleep was 64 bpm. Awake heart rate averaged 78 bpm until the export ended at 15:52 (Fig. 10); the walk and the 25 minutes after it contain almost all of the 10 minutes above 120 bpm.

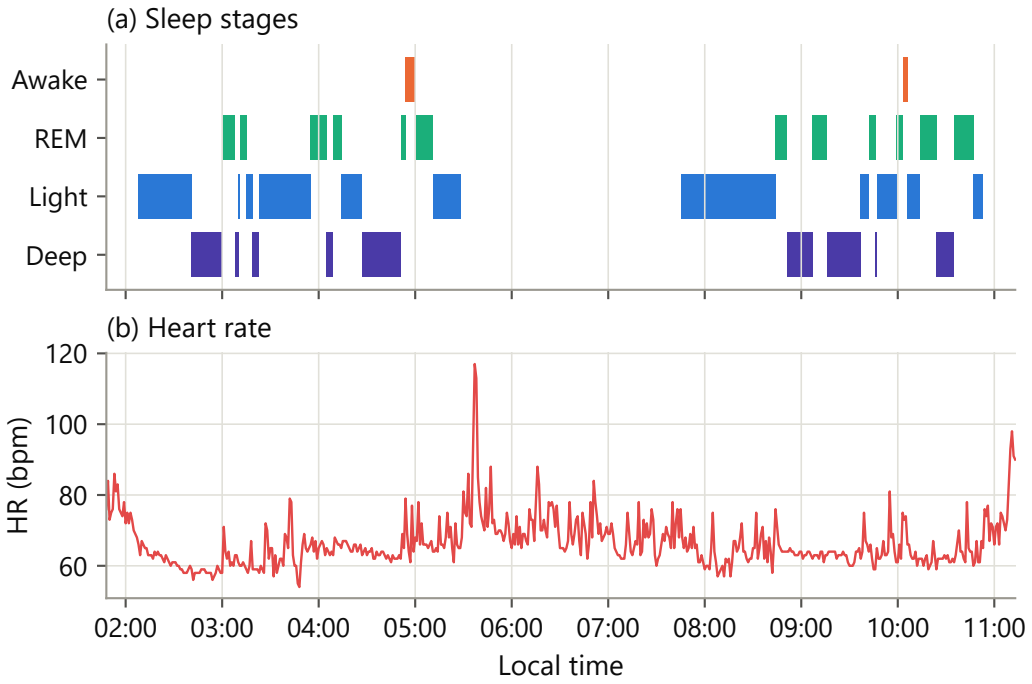


Figure 9: Sleep before day 8 from the wrist band. (a) Hypnogram, 02:08–10:53: 6.3 h asleep (efficiency 98 %), 50 % light, 27 % deep and 23 % REM. (b) Heart rate (mean 64 bpm, lowest 10-min mean 58 bpm). SpO_2 averaged 96.9 % (minimum 93 %).

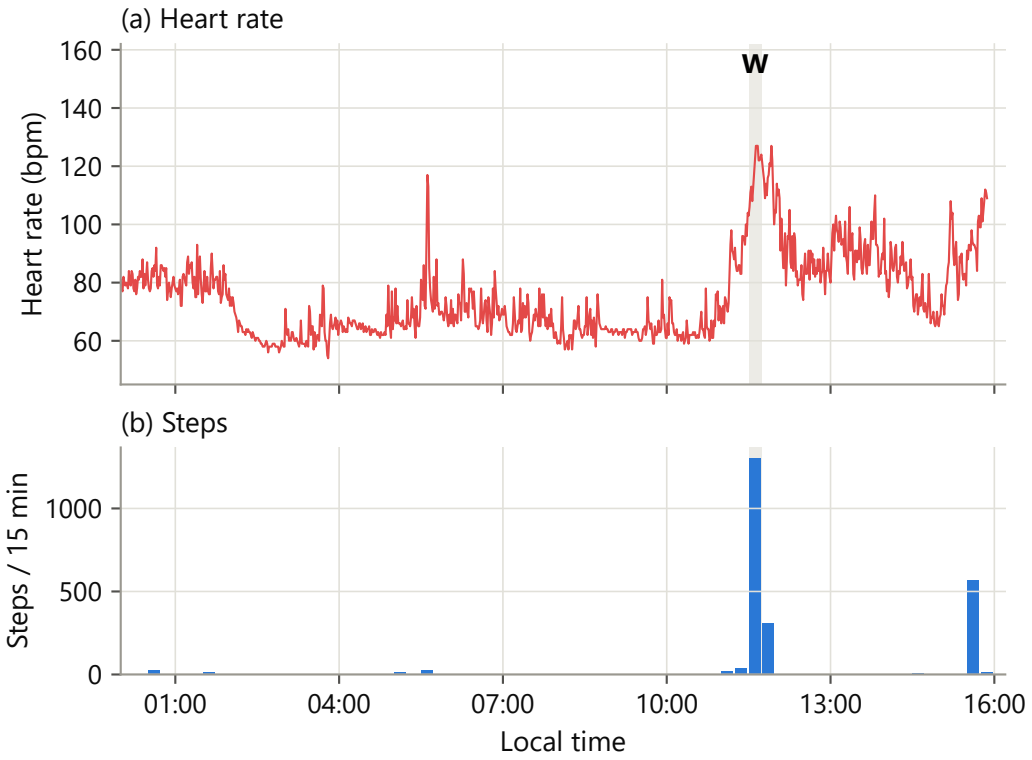


Figure 10: Wrist band over day 8: (a) heart rate (1 sample/min) and (b) steps per 15 min; sleep 02:08–10:53. Grey bands mark the sensor sessions: W morning walk after drinking 500 ml (11:30–11:43).

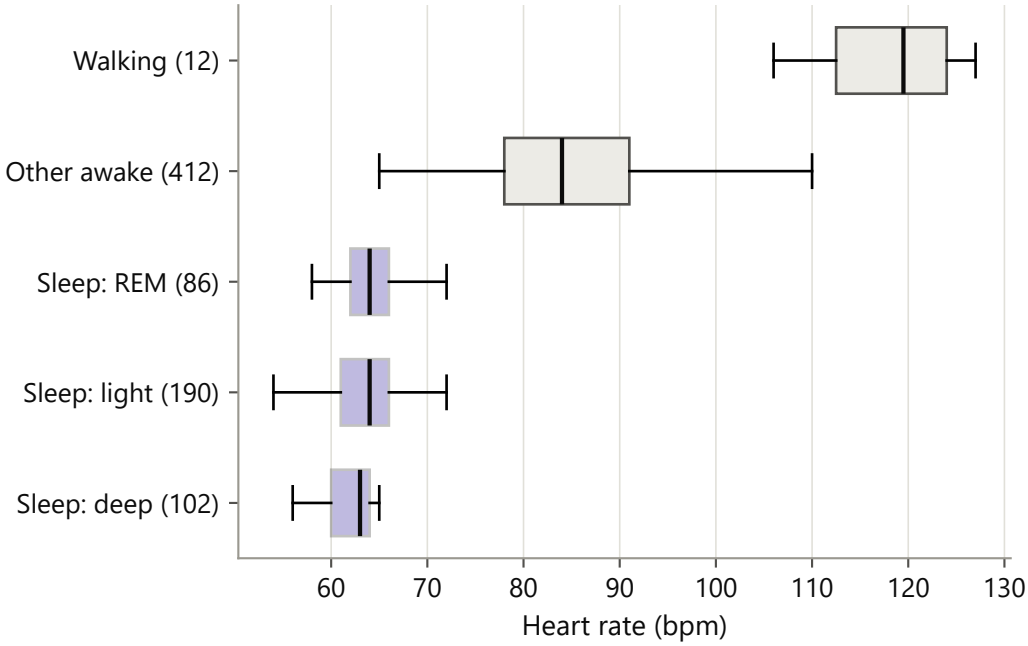


Figure 11: Heart rate by context over the day (minutes per context in brackets). Sleep stages come from the wrist band; awake contexts come from the sensor sessions (gait tests, indoor sitting with the environment node, walking sessions) and all remaining awake minutes. Boxes show the inter-quartile range and median; whiskers extend to 1.5 IQR.

6 Does drinking before walking lower walking heart rate?

On day 7 the evening walk after 500 mL of water had the lowest walking heart rate of the study (92 bpm), which suggested that drinking beforehand might lower heart rate. Day 8 repeated the test on a morning walk. All walks of days 1–8 with a valid walking heart rate were analysed together (11 walks; the day-3 morning walk and the day-6 walk to the metro are excluded because most of their walking minutes were suspected cadence lock). Four walks followed at least 200 mL in the previous 2 hours: day 6 W and H (evening, 300–700 mL), day 7 N (evening, 500 mL) and day 8 W (morning, 500 mL).

Table 3: Walks that followed a drink, against the participant's other walks.

Walk	Start	Water, 2 h	Rest HR	Walk HR	Rise
Day 6 W	21:08	700 mL	133	135	+2
Day 6 H	21:40	300 mL	139	147	+9
Day 7 N	19:16	500 mL	101	92	−9
Day 8 W	11:32	500 mL	86	123	+37
Seven other walks	10:00–19:19	0–160 mL	81–122	107–131	+9 to +36

- *Walking heart rate* was not related to the water drunk in the 2 hours before ($\rho = 0.20$, CI −0.56 to 0.79), to the time since the last drink ($\rho = -0.19$) or to temperature ($\rho = 0.08$). Allowing for temperature, each 100 mL changed walking heart rate by −0.1 bpm (CI −5.4 to +4.7): no detectable effect. The walks after a drink had a median walking heart rate of 129 bpm, the others 126 bpm.
- *Rise above rest.* Up to day 7 the rise looked smaller after drinking ($\rho = -0.61$). Day 8 breaks this pattern: after 500 mL the rise was +37 bpm, the largest of all walks (Fig. 12). With day 8 the association weakens to $\rho = -0.31$ (CI −0.86 to 0.57), and after allowing for the time of day it disappears (partial $\rho = +0.04$).
- *What the earlier pattern really was.* The rise above rest is strongly related to the time of day ($\rho = -0.69$, CI −0.87 to −0.24). Evening walks start from a resting heart rate that is already high (100–139 bpm), so they rise little, and by the evening the participant has also drunk more. The three earlier "drink-before" walks were all in the evening; day 8 is the first in the morning, and it behaves like the other morning walks.

Conclusion. Day 8 does not strengthen the argument; it weakens it. Across 11 walks, drinking before walking had no measurable effect on walking heart rate. The low heart rate of the day-7 evening walk is better explained by that walk's low starting point and the evening timing than by the water. A convincing test would alternate "500 mL" and "no drink" before the same route at the same time of day on several days.

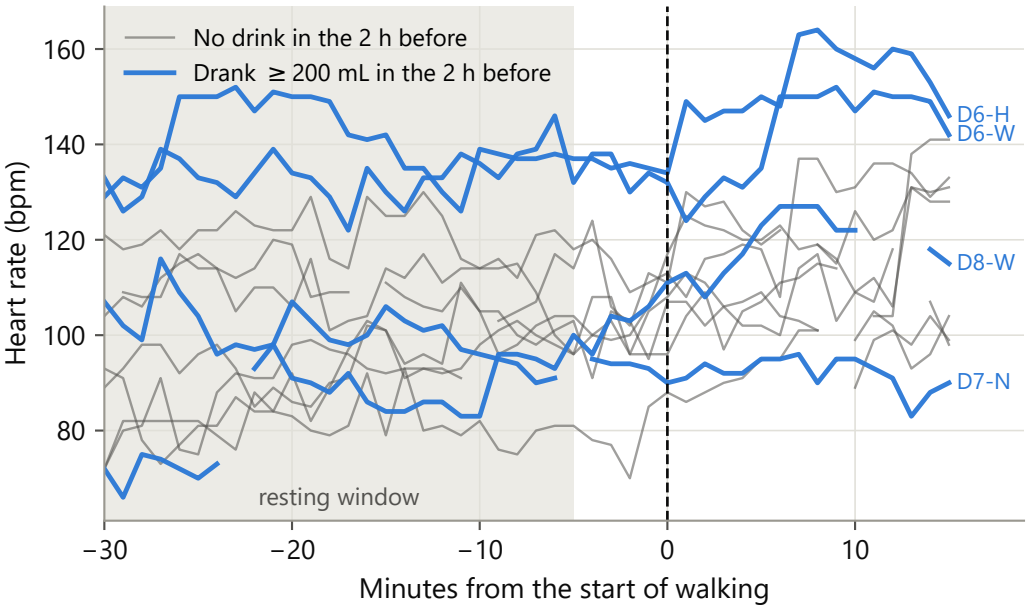


Figure 12: Wrist heart rate around the start of every walk with a valid walking heart rate (11 walks, days 1–8), aligned at the first walking minute (dashed). Blue: walks with at least 200 mL drunk in the previous 2 hours (labelled); grey: the others. The shaded band is the resting window used for the rise of heart rate above rest.

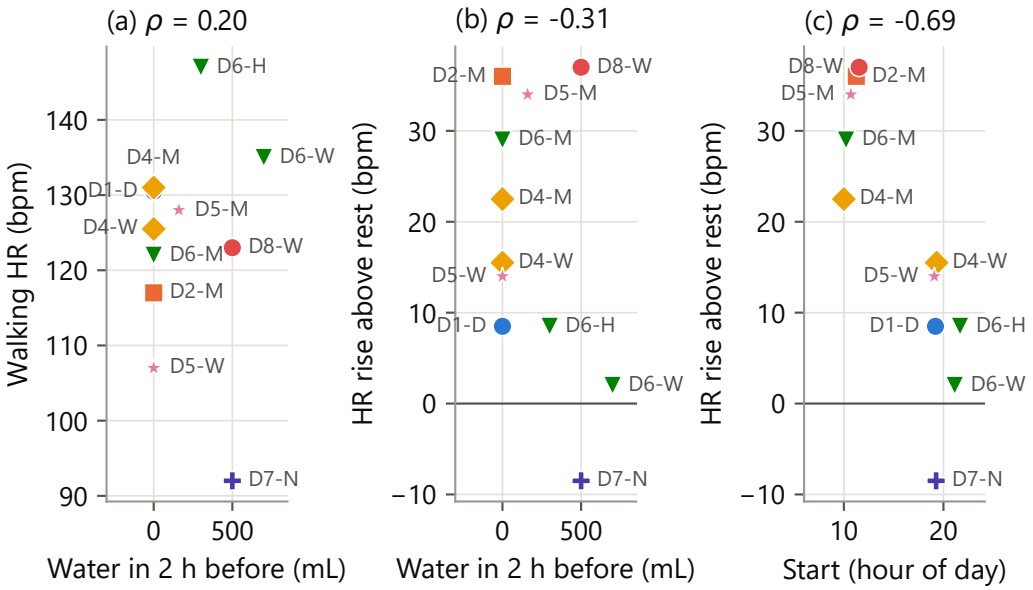


Figure 13: Water drunk before walking and heart rate during the walk (11 walks over days 1–8; median heart rate of walking minutes excluding suspected cadence lock). Walks D3-M, D6-E are excluded because most of their walking minutes were suspected cadence lock. (a) Walking heart rate against fluid from the bottle log in the 2 hours before the walk; (b) rise of heart rate above the resting value 5–30 minutes before the walk, against the same; (c) that rise against the time of day. Spearman ρ with bootstrap 95 % CI: (a) 0.20 (-0.58 to 0.79), (b) -0.31 (-0.86 to 0.60), (c) -0.69 (-0.86 to -0.23); (b) after allowing for the time of day: partial $\rho = 0.04$. Labels: day and walk.

7 Comparison of days 1–8

Day 8's walk was the fastest of the study (cadence 108–109 steps/min, speed 1.25 m/s), while stance (62–63 % on both feet), push-off timing and step timing stayed within the ranges of the previous seven days (Figs. 14 and 15). Sleep (6.3 h) and heart rate during sleep (64 bpm) were typical.

Table 4: Headline metrics across days 1–8 (medians for environment values; left lag: left heel strike as % of the right gait cycle; †walking heart rate excludes minutes of suspected optical cadence lock, i.e. heart rate within 6 bpm of step rate for ≥2 consecutive minutes; stance and swing from IMU gait events; HR-lock min.: flagged / walking minutes).

Metric	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8
Time asleep (h)	6.6	6.3	6.6	6.1	7.5	7.4	5.4	6.3
Deep sleep (%)	27	36	35	28	28	32	30	27
REM sleep (%)	31	19	19	19	33	28	23	23
Mean HR asleep (bpm)	60	60	61	64	63	63	64	64
Mean awake HR (bpm)	94	88	92	98	96	104	96	86
HR > 120 bpm (min)	56	14	5	66	56	241	24	10
Steps (band)	7,173	7,044	5,661	7,021	4,863	8,641	4,284	2,347
Steps (phone)	8,109	6,608	5,422	8,492	5,131	8,435	4,694	–
HR–steps ρ	0.27	0.33	0.23	0.30	0.26	0.27	0.08	0.34
Walk cadence (steps/min)	105	101	102	100	97	102	100	109
Stride time (s)	1.14	1.19	1.18	1.19	1.24	1.17	1.20	1.10
Stride-time CV (%)	10.6	6.7	12.6	12.1	9.7	5.4	15.4	7.9
Stance R (%)	63	63	62	60	63	62	63	63
Stance L (%)	–	–	63	64	63	62	62	62
Swing R (s)	0.42	0.43	0.44	0.48	0.46	0.45	0.44	0.42
Swing L (s)	–	–	0.45	0.43	0.46	0.44	0.47	0.42
Push-off timing R (%)	66	66	66	62	66	66	66	65
Push-off timing L (%)	–	–	67	66	67	66	67	66
Upright sitting (%)	14	–	–	54	9	6	18	–
Walk distance (m)	925	1,824	1,449	1,290	917	624	747	741
Walk speed (m/s)	1.19	1.13	1.13	1.10	1.06	1.15	1.18	1.25
Stride length (m)	1.34	1.32	1.32	1.29	1.30	1.32	1.27	1.37
Walk mean HR† (bpm)	124	111	–	119	113	–	91	120
HR-lock min.†	0	0	22	5	3	7	3	3
Walking min.	16	35	30	17	20	11	10	10
Left lag (% cycle)	49.7	–	49.2	49.7	49.2	49.3	48.8	49.5
Indoor $PM_{2.5}$ ($\mu g/m^3$)	51	67	–	48	52	51	40	–
Indoor temperature (°C)	21.2	29.5	–	23.5	26.1	24.9	26.9	–
Indoor sound (dBA)	67	56	–	65	63	72	56	–
Outdoor $PM_{2.5}$ ($\mu g/m^3$)	59	71	101	68	68	71	57	59
Bedroom $PM_{2.5}$ ($\mu g/m^3$)	–	84	69	102	75	76	60	–
Fluid intake (mL)	1,744	1,292	2,158	1,660	890	1,967	1,265	–
Longest drink gap (h)	6.0	5.1	3.0	9.5	9.4	9.8	7.6	–

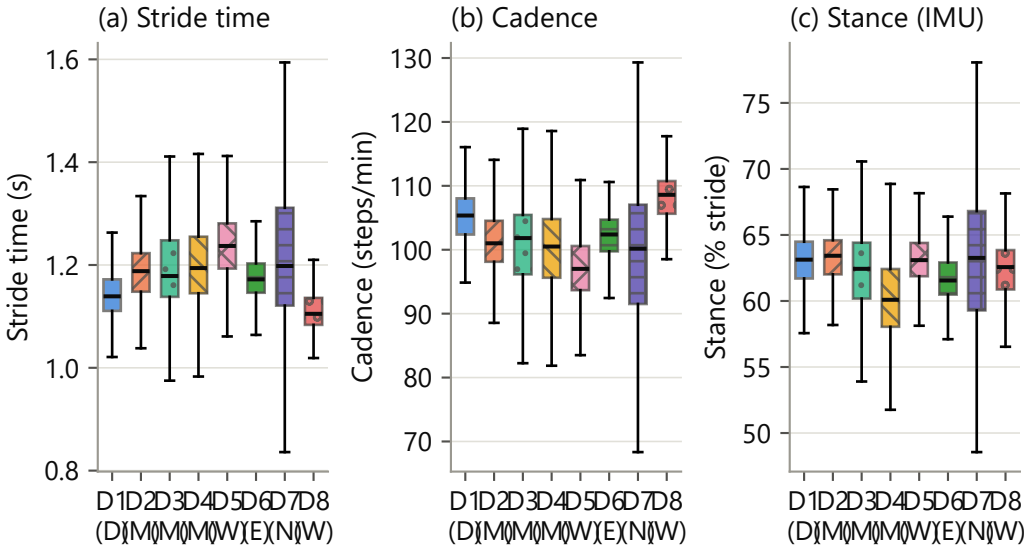


Figure 14: Right-foot gait during the main walk of each day (steady walking only: bouts of at least 10 strides without their first and last two strides; day 1: session D, 896 steady strides; day 2: session M, 1842 steady strides; day 3: session M, 1560 steady strides; day 4: session M, 912 steady strides; day 5: session W, 983 steady strides; day 6: session E, 599 steady strides; day 7: session N, 656 steady strides; day 8: session W, 615 steady strides). Cadence is 120 divided by stride time. Stance is measured from IMU gait events (heel-strike impact to IMU toe-off), which stay valid after the right first-metatarsal pressure sensor was replaced (day 5 onwards).

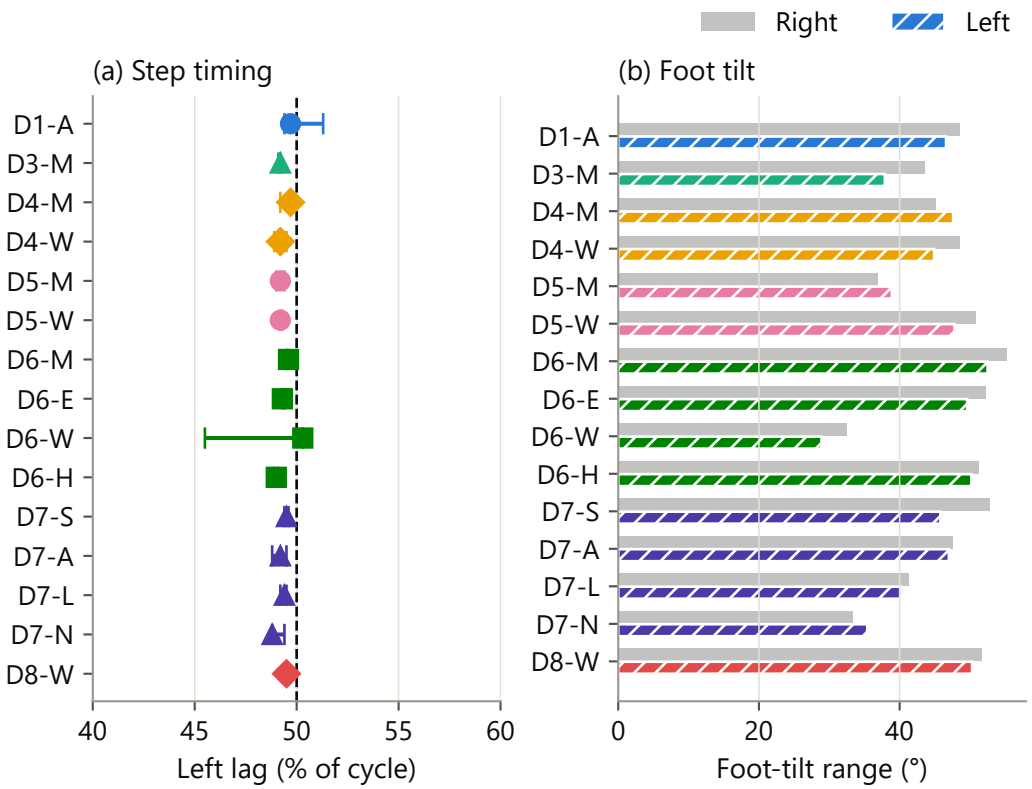


Figure 15: Left/right comparison in every session where the left insole delivered enough data (labels: day and session). (a) Lag of the left heel-strike impact behind the right one, as a percentage of the gait cycle, with bootstrap 95 % confidence intervals; 50 % (dashed) is perfectly symmetric step timing. (b) Range of foot tilt over the cycle for each foot. Sessions where the left and right acceleration curves did not match in shape are omitted.

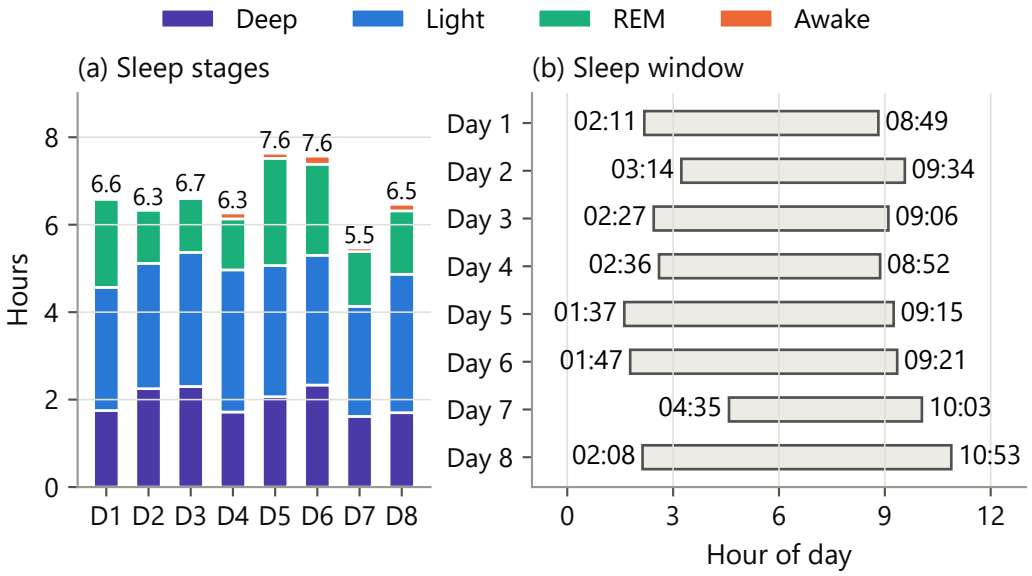


Figure 16: Sleep on the night before each day (wrist band): (a) time in each stage and total; (b) sleep onset and wake time.

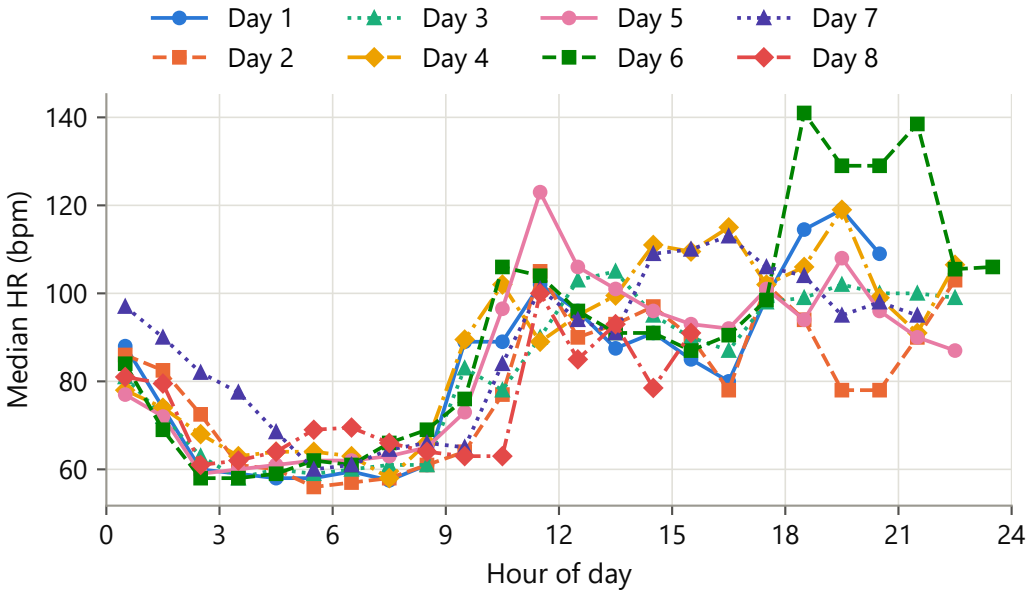


Figure 17: Hour-by-hour median heart rate (wrist band) on each day. Hours without band data are not drawn.

8 Heart rate and step rate over eight days

The relationship between wrist heart rate and step rate was positive on all eight days ($\rho = 0.08$–0.33; day 8 0.29; two-sided sign test $p = 0.008$), with a pooled within-day $r = 0.30$ (95 % CI 0.24–0.35). Leaving out any one day gives 0.28–0.32, and removing the 45 suspected cadence-lock minutes leaves it at 0.30. It explains about 9 % of the variation in heart rate (20 bpm per 100 steps/min) and appears only over windows of 10 minutes or more (Fig. 18).

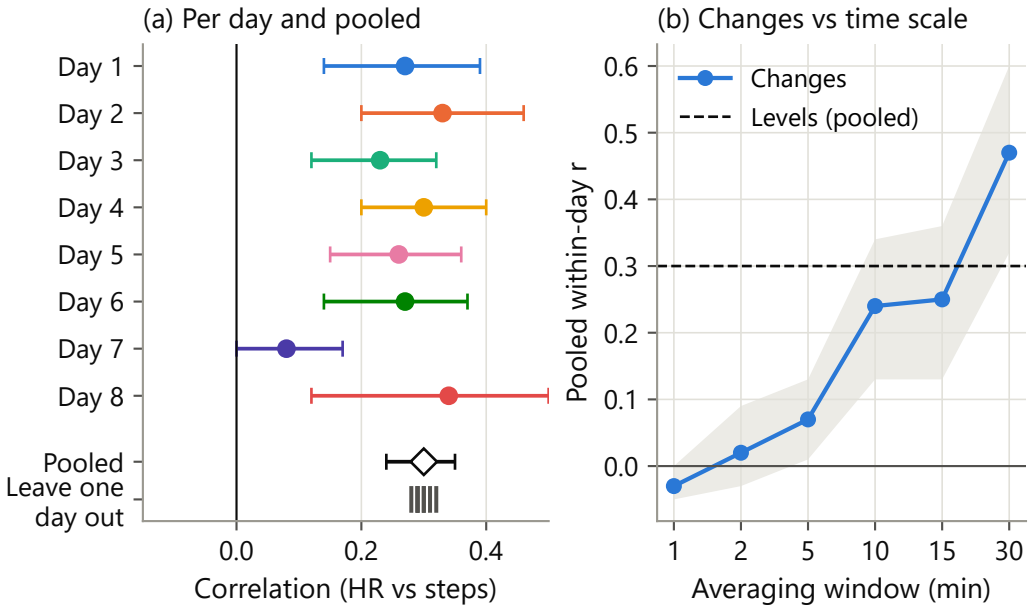


Figure 18: Robustness of the relationship between wrist heart rate and step rate in awake minutes (6,792 minutes, days 1–8). (a) Spearman ρ per day with 10-minute block-bootstrap 95 % CIs; pooled within-day Pearson r (diamond) and the pooled r when each day in turn is left out (ticks: 0.28–0.32). (b) Pooled within-day correlation between *changes* of heart rate and step rate after averaging over windows of 1 to 30 minutes (band: 95 % CI); dashed: correlation of levels. Minute-to-minute changes are unrelated ($r = -0.03$) because heart rate responds to activity over several minutes; over 30-minute windows the changes correlate with $r = 0.47$.

Table 5: Relationships between per-minute measurements, days 1–8. Pairs are listed as heart rate (or the first variable named) against the second. Per day: Spearman ρ on each day. Level: pooled within-day (repeated-measures) Pearson r with 10-minute block-bootstrap 95 % CI. Change: the same for minute-to-minute changes (*: CI excludes zero). Consistent: same sign with $|\rho| > 0.2$ on every day, both CIs excluding zero and $|r| \geq 0.3$; level only: consistent in levels but not in changes; varies: pooled CI excludes zero but per-day signs or sizes differ. Covariances and regression slopes are given in relationships.csv.

	Per-day ρ										
Pair	D1	D2	D3	D4	D5	D6	D7	D8	Level r (CI)	Change	Verdict
Heart rate, awake											
Steps	.27	.33	.23	.30	.26	.27	.08	.34	.30 (.24, .35)	−.03*	varies
Heart rate, walking											
Cadence	.16	.00	–	.19	.25	−.08	.07	–	.09 (−.10, .25)	−.02	none
Speed	−.10	−.18	–	−.02	.34	.20	.05	–	.19 (−.13, .43)	−.05	none
Temperature	−.78	.59	–	.67	.42	−.52	.34	–	.32 (−.02, .62)	.12	none
$PM_{2.5}$	−.12	−.20	–	−.25	−.33	.37	−.45	–	−.21 (−.35, .00)	−.13	none
Sound	.01	.07	–	−.11	.28	.16	−.46	–	.05 (−.22, .31)	.09	none
Heart rate, asleep											
Temperature	–	.38	.00	.11	.16	.51	.19	–	.24 (.16, .33)	−.03	varies
CO_2	–	−.32	.19	.04	−.11	.02	−.32	–	−.09 (−.17, .00)	−.01	none
$PM_{2.5}$	–	−.23	.12	.38	−.36	−.59	.38	–	−.04 (−.12, .04)	−.01	none
Sound	–	−.32	−.10	.42	−.33	−.45	−.31	–	−.23 (−.32, −.14)	.01	varies
Gait and sensors, walking											
Stride vs cadence	−.29	−.55	−.89	−.68	−.50	−.69	−.38	−.73	−.42 (−.61, −.25)	−.33*	consistent
Insole vs band	.42	.74	.51	.60	.42	.60	.16	.12	.34 (.19, .50)	.05	varies
Environment											
Sound vs $PM_{2.5}$	.23	.60	.72	.31	−.17	.27	.23	.88	.31 (.20, .43)	.00	varies
PM_{10} vs $PM_{2.5}$	.96	1.00	1.00	1.00	1.00	1.00	1.00	.97	1.00 (1.00, 1.00)	.99*	consistent
RH vs temperature	.09	−.93	−.90	−.08	.08	−.64	−.24	−.83	.22 (.11, .32)	−.13*	varies